“Empowerment through quality technical education”
AJEENKYA DY PATIL SCHOOL OF ENGINEERING
Dr. D. Y. Patil Knowledge City, Charholi Bk., Via. Lohegaon, Pune – 411081
Imparting quality education in the field of Artificial Intelligence and Data science.
Department of Artificial Intelligence and Data Science Engineering

ACADEMIC YEAR: 2026-27	
SUBJECT: PCC302AID:Machine Learning	
CLASS: TE-	**SEMESTER:** V
CASE STUDY TITLE: Type your title here	
NAME OF STUDENT:	
ROLL NO:	
DATE OF SUBMISSION:	
NAME AND SIGN OF SUBJECT INCHARGE: Prof.	

CASE STUDY FORMAT – NLP

Use the following **standard format** for preparing NLP case studies:

1. Title of the Case Study

Develop and Analyze a Part-of-Speech (POS) Tagging System Using a Probabilistic Sequence Model

2. Introduction

Brief introduction to the NLP concept and its importance.

3. Problem Statement

Clearly define the real-world problem that needs to be solved.

4. Case Study Scenario

Describe the real-world situation/application where the NLP technique is used.

5. Objectives

List **4–6 specific objectives** of the case study.

6. Dataset Description

Mention:

- Dataset name/source
- Number/type of records
- Input features
- Output/target
- Training and testing data

7. Methodology / Proposed Approach

Explain the steps followed to solve the problem.

Example:

```
Dataset → Preprocessing → Feature Extraction → Probabilistic Model → Prediction → Evaluation
```

8. NLP Techniques / Algorithms Used

Mention the relevant techniques and algorithms.

For POS tagging:

- Tokenization
- POS tagging
- Hidden Markov Model (HMM)
- Transition Probability
- Emission Probability
- Viterbi Algorithm

9. System / Model Development

Explain how the proposed NLP system is developed.

Include:

- Input
- Processing
- Model
- Output

10. Working / Example

Provide one practical example showing:

Input Sentence → Processing → Predicted POS Tags

11. Results and Analysis

Present the obtained results using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Also discuss the observations.

12. Challenges / Limitations

Discuss problems such as:

- Ambiguous words
- Unknown words
- Limited training data
- Context dependency
- Domain-specific vocabulary

13. Applications

Mention real-world applications of the developed NLP system.

14. Advantages

List the benefits of the proposed approach.

15. Future Scope

Explain how the system can be improved using advanced techniques such as:

- CRF
- BiLSTM
- BERT
- Transformer models

16. Learning Outcomes

Students should be able to:

- Understand the NLP problem.
- Apply appropriate NLP techniques.
- Implement the selected algorithm.
- Analyze model performance.
- Identify practical challenges.

17. Conclusion

Summarize the case study, methodology, results, and practical significance.

CO Mapping:
PO Mapping:
SDG Mapping:

IMPORTANT: Since this is a **descriptive case study**, make sure that: